

RN.SONiC.202111.2 2022 Nov

Enterprise SONiC Distribution by Edgecore

Release Date: 2022/11/25

Version:

- Broadcom Trident 2+ / Trident 3 / Tomahawk / Tomahawk 3 Series
 - Edgecore-SONiC_20221117_052015_ec202111_245
- Broadcom Trident 4 / Tomahawk 4 Series
 - Edgecore-SONiC_20221117_052611_ec202111_hsdk_6.5.23_192
- Intel Tofino / Tofino 2 Series
 - Edgecore-SONiC_20221117_051600_ec202111_bfn_212
- Virtual Switch
 - Edgecore-SONiC_20221117_131345_ec202111_vs_47

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Overview

The Edgecore SONiC 202111.2 update primarily supports the hardware platforms and new features over the 202111.1 version.

Supported hardware platforms

- AS9736-64D (Broadcom Tomahawk 4), verified on hardware revision ROC
- DCS501 / AS7712-32X

Features enhanced

- Static LAG
- Auto-negotiation for Clause 37 and Clause 73

Platform Supported

Broadcom Trident 2+ / Trident 3 / Tomahawk / Tomahawk 3 Series

- EPS201 / AS4630-54TE
- EPS202 / AS4630-54PE
- EPS203 / AS4630-54NPE
- DCS201 / AS5835-54X
- DCS202 / AS5835-54T
- DCS203 / AS7326-56X
- DCS204 / AS7726-32X
- DCS208 / AS5812-54X
- DCS501 / AS7712-32X
- DCS510 / AS9716-32D

Broadcom Trident 4 Series

- DCS240 / AS9726-32DB

Broadcom Tomahawk 4 Series

- AS9736-64D

Intel Tofino / Tofino 2 Series

- DCS800 / Wedge100BF-32X
- DCS801 / Wedge100BF-32QS
- DCS802 / Wedge100BF-65X
- DCS810 / AS9516-32D

Notes for Design Changes from 202012 to 202111

The following are the notes for the behavior changes between the 202012 and 202111 releases (same as the 202111.1 release).

Feature	Differences				
MCLAG	(1) Unique IP setting is an essential configuration in 202111. Need to manually add unique IP setting if config_db.json is migrated from 202012. • 202012: Unique IP is optional • 202111: Unique IP is mandatory (2) There are restrictions on the CLI before enabling/disabling unique IP. Please refer to the CLI guide for more details ("config mclag unique-ip add/del").				
BGP Unnumbered	"link-local-only" mode is introduced in 202111 for directly enabling BGP unnumbered on the interface without IP settings. • 202012: The ethernet interface didn't require the additional setting for BGP unnumbered • 202111: For ethernet interface without any IP configuration, "link-local-only" mode must be enabled first before using BGP unnumbered (config interface ipv6 enable use-link-local-only <interface_name>).				
Interface Counters	The definition of RX_BPS, RX_UTIL, TX_BPS, and TX_UTIL counters are changed in 202111. • 202012: The average rate/utilization in the specified time • 202111: The exponential moving average rate/utilization after the system starts up				
QoS	The valid PG range is changed in Intel TF/TF2 platforms. • 202012: 0 ~ 4 • 202111: 1 ~ 5				
QoS	QoS related table in config_db.json has been updated in the 202111 release (ABNF format is canceled). Example: <table> <tr> <th>202012</th><th>202111</th></tr> <tr> <td> <pre>"Ethernet92 3": { "scheduler": "[SCHEDULER scheduler.1]", "wred_profile": "[WRED_PROFILE AZURE_LOSSLESS]" }, "Ethernet0 0": { "profile": "[BUFFER_PROFILE ingress_lossy_profile]" }, "Ethernet0": { "dscp_to_tc_map": "[DSCP_TO_TC_MAP AZURE]", "pfc_to_queue_map": "[MAP_PFC_PRIORITY_TO_QUEUE AZURE]", "tc_to_pg_map": "[TC_TO_PRIORITY_GROUP_MAP AZURE]", "tc_to_queue_map": "[TC_TO_QUEUE_MAP AZURE]" },</pre> </td><td> <pre>"Ethernet92 3": { "scheduler": "scheduler.1", "wred_profile": "AZURE_LOSSLESS" }, "Ethernet0 0": { "profile": "ingress_lossy_profile" }, "Ethernet0": { "dscp_to_tc_map": "AZURE", "pfc_to_queue_map": "AZURE", "tc_to_pg_map": "AZURE", "tc_to_queue_map": "AZURE" },</pre> </td></tr> </table>	202012	202111	<pre>"Ethernet92 3": { "scheduler": "[SCHEDULER scheduler.1]", "wred_profile": "[WRED_PROFILE AZURE_LOSSLESS]" }, "Ethernet0 0": { "profile": "[BUFFER_PROFILE ingress_lossy_profile]" }, "Ethernet0": { "dscp_to_tc_map": "[DSCP_TO_TC_MAP AZURE]", "pfc_to_queue_map": "[MAP_PFC_PRIORITY_TO_QUEUE AZURE]", "tc_to_pg_map": "[TC_TO_PRIORITY_GROUP_MAP AZURE]", "tc_to_queue_map": "[TC_TO_QUEUE_MAP AZURE]" },</pre>	<pre>"Ethernet92 3": { "scheduler": "scheduler.1", "wred_profile": "AZURE_LOSSLESS" }, "Ethernet0 0": { "profile": "ingress_lossy_profile" }, "Ethernet0": { "dscp_to_tc_map": "AZURE", "pfc_to_queue_map": "AZURE", "tc_to_pg_map": "AZURE", "tc_to_queue_map": "AZURE" },</pre>
202012	202111				
<pre>"Ethernet92 3": { "scheduler": "[SCHEDULER scheduler.1]", "wred_profile": "[WRED_PROFILE AZURE_LOSSLESS]" }, "Ethernet0 0": { "profile": "[BUFFER_PROFILE ingress_lossy_profile]" }, "Ethernet0": { "dscp_to_tc_map": "[DSCP_TO_TC_MAP AZURE]", "pfc_to_queue_map": "[MAP_PFC_PRIORITY_TO_QUEUE AZURE]", "tc_to_pg_map": "[TC_TO_PRIORITY_GROUP_MAP AZURE]", "tc_to_queue_map": "[TC_TO_QUEUE_MAP AZURE]" },</pre>	<pre>"Ethernet92 3": { "scheduler": "scheduler.1", "wred_profile": "AZURE_LOSSLESS" }, "Ethernet0 0": { "profile": "ingress_lossy_profile" }, "Ethernet0": { "dscp_to_tc_map": "AZURE", "pfc_to_queue_map": "AZURE", "tc_to_pg_map": "AZURE", "tc_to_queue_map": "AZURE" },</pre>				

Feature Enhancements

RN.Enhancement

Support static LAG feature (config portchannel add <portchannel_name> [--static {true | false}]).

Support the link-training configuration CLI command (config interface link-training <interface_name> <mode: on | off>).

Enhance Port Channel to support LACP rate configuration (config portchannel add <portchannel_name> [--fast-rate {true | false}]).

Support 4x100G application mode on ET7502-DR4 transceiver.

[Broadcom TD3] Support DSCP/PCP remarking.

Enlarge the quantity of VRF from 1000 to 4096.

Support CL73 auto negotiation.

Support CPU and MAC chip temperature retrieval.

Support MIB ifPhysAddress (1.3.6.1.2.1.2.2.1.6).

Enhance ACL loader to support Source/Destination IP and VLAN as the classification match condition.

Support the source IP specification for syslog outgoing packets.

Support FRR reload script in FRR BGP container.

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Fixed Issues

TC Number	RN.Enhancement
SONIC#5375	ARP packets from the leaf nodes to the spine aren't encapsulated in VxLAN tunnel.
SONIC#4321	[AS4630-54PE/TE/NPE][QoS] Fix the problem that the traffic forwarding is still restricted after removing the shaper from the egress queue.
SONIC#4851	[AS9726-32DB] Fix the packet loss issue when shutting down and starting up the interface between leaf & spine switches.
SONIC#5329 (#24685 /#24989)	Fix the login failure after using ZTP to install the new image.
JRTC#10796	Fix the link-flap issue during fast reboot.
JRTC#10845	Fix the packet loss issue during warm reboot.
JRTC#11071	[MCLAG] Fix the problem that the MAC addresses failed to sync to peer device when MCLAG member port is down.
SONIC#3872	[Broadcom TD2+/TD3/TH3/TD4][Egress Port Shaping] Fix the problem that egress port rate-limit limits the bandwidth on each queue, not per-port basis.
SONIC#4541	[Broadcom TD2+/TD3/TH3/TD4] Fix the ACL type missing problem (MIRROR_DSCP) in ACL table creation CLI command.
SONIC#4683	[Intel TF/TF2] Fix the SWSS and SyncD container crash problem when executing Fast Reboot and Warm Reboot in sequence.
SONIC#4704 (#23307)	[MCLAG] Fix the FDB wrong learning issue when enabling unique IP and rebooting the device repeatedly.
SONIC#4732 (#23554)	[MCLAG] Fix the FDB wrong learning to the device own MAC address when rebooting the device repeatedly.
SONIC#4749	[AS9726-32DB] Fix the SyncD crash problem when executing "show interfaces status" after container warm restart.

SONIC#4338 & SONIC#4767	Fix the limitation that RADIUS authorization needs a user account created on switch first before use it.
SONIC#4794	[AS9726-32DB] Fix the VxLAN traffic loss issue in MLAG environment.
SONIC#4844	[AS4630-54TE] Fix the transceiver EEPROM reading issue on Compliance Code & DOM Capability.
SONIC#4861	[Intel TF/TF2] Fix the MAC learning problem after clearing all FDB entries with command "sonic-clear fdb all".
SONIC#4874 (#23450)	[AS9716-32D] Force the WAN PLL to free run mode by default.
SONIC#4880 (#25057)	Fix the EVPN-MH related FRR configuration missing problem after system rebooting.
SONIC#4914 (#23405 /#24428)	Fix the error when running DHCP Client on switch ports.
SONIC#4917	[AS5812-54X] Fix the EEPROM display problem after moving the 10G DAC cable to the new port.
SONIC#4918 (#23953)	Fix the VRF route entry missing problem after system reboot or reload.
SONIC#4921	[RADIUS] Fix NAS IP configuration not working problem in CLI.
SONIC#4922	[RADIUS] Fix the problem that Source IP and source interface configuration in CLI are not effective.
SONIC#4938 (#24007)	Fix the inconsistent route table issue when using BGP unnumbered with IPv6 link local.
SONIC#4944 & SONIC#4945 (#24035/#24775)	[AS4630-54TE] Fix the fan direction display issue to show "exhaust" or "intake" correctly.
SONIC#5000 (#23681)	Fix the Diag LED wrong indication problem caused by system health.
SONIC#5012	Fix the problem that the static route in MLAG and SAG use case is not preserved after system rebooting.
SONIC#5114 (#24209)	Enhance Xcvrd to set the appropriate pre-emphasis Credo AEC cables.
SONIC#5139 (#24289)	[AS5812-54X] Fix the problem that the default Auto-Nego mode is displayed as "N/A".
SONIC#5159 (#24230 /#24310)	Fix the system crash problem when setting FEC on the interface with auto-nego enabled.
SONIC#5175/#5159/#4870 (#24230)	Enhance SONiC orchagent to avoid the container crash when the users configure the incompatible FEC mode and try to apply the FEC mode again when the port speed is changed.
SONIC#5211 (#24386)	[TACACS] Fix the device abnormal busy and slow response issue when there is a TACACS server offline in the authentication list.
SONIC#5212 (#24423 /#24904)	Fix the system crash problem when changing MTU on Port Channel.
SONIC#5249 (#24504)	Enhance TACACS CLI to support global source IP address configuration.
SONIC#5251 (#24503)	Fix the wrong syslog message when TACACS authentication passed.
SONIC#5264	Fix the pre-emphasis update problem when the transceiver is unplugged and re-plugged after Fast Reboot.
SONIC#5275 (#24567)	[MLAG] Fix the potential system crash problem when removing a Port Channel from VLAN member and it is also a member of MLAG.
SONIC#5277	Fix the system crash problem when the VxLAN tunnel port is a member of Port Channel and it also joined a VLAN with IP address.
SONIC#5289	Fix FRR crash problem when executing "show bgp all" and "show bgp vrf all all" in FRR.
SONIC#5301 (#24653 /#24680)	Fix the problem that the route entries are not synced to switch ASIC when using FRR to configure static route leaks between the different VRFs.
SONIC#5320	Fix the EVPN-MH handover and FRR MH configuration missing problem when rebooting the leaf switches.
SONIC#5331 (#24684)	Fix the obsolete data issue in sub-port related REST API after removing the sub-port.
SONIC#5340 (#24691)	Fix the SyncD container stuck problem when the port interface stays in link flapping.
SONIC#5341	Fix the compliance code display problem in 10G transceiver.
SONIC#5359	[Intel TF/TF2] Fix the problem that egress ACL cannot be created for mirror session.

SONIC#5418 (#24873)	Fix the problem that "mux" section in config_db.json will be generated with invalid settings when saving the configuration before the system is not yet ready.
SONIC#5435 (#24904 /#25206)	Fix the container crash problem when the same physical port to create multiple sub-interfaces are created on the same physical port and and bound to the different VRFs.
SONIC#5464 (#25030)	Deny the IP configuration on the member port of Port Channel.
SONIC#5479	[AS9716-32D] Support low power mode set/get function for QSFP-DD.
SONIC#5593	Fix the incorrect L2 & L3 VNI list problem in FRR EVPN show command.

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Known Issues

TC Number	RN.Description
SONIC#5515	[AS9736-64D (ROC)] NTP cannot successfully sync the time form server.
SONIC#4320	[AS4630-54PE/TE/NPE][PFC] PFC WD & PFC counters are not working on 100G ports.
SONIC#5168	[Broadcom TH4] CRM cannot provide the correct usage of ACL table/group resource.
SONIC#4601	[QoS] WRED behavior will become abnormal after binding new profile to the port interface (old and new profile settings would be mixed together).
JRTC#5469	[Intel TF/TF2] Warm reboot time may exceed 90 secs and cause LAG link flapping because of LACP timeout.
SONIC#4134	[Intel TF/TF2] The port interface might not be link-up when the fiber transceiver is back-to-back removed and inserted (require 2 secs delay between pull-out and plug-in operations).
SONIC#4146	[Intel TF/TF2] The EEPROM data of fiber type transceiver/cable cannot be displayed.
SONIC#4203	[Intel TF/TF2] QSFP28 transceiver/cable EEPROM data cannot be correctly displayed.
SONIC#4326	[NAT] SONiC container crash occurs after disabling NAT feature ("config feature state nat disabled").
SONIC#4607	[AS9726-32DB][PFC WM] Priority group watermark counters are not working.
SONIC#4608	[AS9726-32DB][PFC WM] Egress unicast queue watermark counters are not working.
SONIC#4857	[Intel TF/TF2] The switch would no longer be able to learn the new MAC address when the FDB clearing and new MAC learning occurred concurrently.
SONIC#4858	[Intel TF/TF2] QoS re-marking for 802.1p is not working.
SONIC#4867	[Intel TF/TF2] Policer cannot be used to perform the rate limit for mirror packets.
SONIC#4916	CLI command "config load" is not working and will cause SWSS crash if there is a configuration change on the buffer profile.
SONIC#5539	System crash occurs when configuring packet mode storm control with CBS argument.

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Limitations

TC Number	RN.Description
SONIC#043 2	[Broadcom TD2+/TD3/TH3/TD4][ECN] ECN only support on (ecn_all, ecn_red, ecn_green_red, ecn_yellow_red) to enable the ECN functionality. For other options, it does not take effect.
SONIC#037 5	[Broadcom TD2+/TD3/TH3/TD4][PFC] PFC WD only can apply to 1 or 2 queues.
SONIC#406 8	[AS9716-32D] PFC WM headroom counters are not supported, "N/A" is shown in following commands. "show headroom-pool watermark"
SONIC#407 4	[Intel TF/TF2] PFC WM headroom/shared counters are not supported. "N/A" is shown in the following commands: "show headroom-pool watermark" "show priority-group watermark headroom" "show priority-group watermark shared"
JRTC#0622	[Broadcom TD2+/TD3/TH3/TD4][Drop Counter] feature "drop counter" only applied on ingress and supports these 6 drop reasons (L3_EGRESS_LINK_DOWN, EXCEEDS_L3_MTU, SMAC_EQUALS_DMAL, ACL_ANY, DIP_LINK_LOCAL, SIP_LINK_LOCAL). It has accuracy issue which introduced by one counter shared by multi-reasons in chipset.
JRTC#1136	[VRF] VRF attributes modification is not supported, which includes IPv4/v6 admin state, source MAC, and violation TTL1/IP options /L3 multicast actions.
JRTC#1571	[AS4630-54PE] All PFC related features (PFC, PFC Watchdog, PFC Watermark) cannot be supported in 1GbE interfaces.
SONIC#419 2	[Intel TF/TF2] No ECMP support for overlay BUM traffic.
SONIC#419 4	[Intel TF/TF2][EVLAN/VxLAN] VxLAN encapsulation is not working in asymmetric route.
SONIC#475 0	[Intel TF/TF2] Twice NAT performance is limited due to a lack of hardware-based translation support, only software-based translation is available.
SONIC#477 7	[Intel TF/TF2] PFC to Queue mapping is not changeable and the default mapping are 0:0, 1:1, 2:2, ... 7:7.
SONIC#480 6	[Intel TF/TF2] the behavior of XON is the same as XON-Offset in buffer profile.
SONIC#484 6	[Intel TF/TF2] PG watermark in "show priority-group watermark shared" includes not only the shared size but also the headroom size.
SONIC#486 6	[Intel TF] Mirroring the specific DSCP packet is not supported.
SONIC#554 2	[Broadcom] No traffic rate limit on port mirroring.
SONIC#555 2	[Intel TF/TF2] Policer doesn' t support bytes mode on X2/Y2 profile.

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